



ALMOST FOSSILIZED

Any evidence of life that survives the processes of decay and is preserved for 10,000 years or more can be called a fossil. Ice-age plant pollen preserved in peat bogs is a type of fossil; however, since it is not turned to stone, it is sometimes called a "subfossil." The same is true of this fly, which was trapped in sticky tree resin about 20 million years ago and was preserved because the resin hardened to form amber.



dinosaur to life.



GRAPHIC EVIDENCE

The delicate tissues of dead organisms usually decay rapidly or are eaten. This means that most fossils are of scattered shells, bones, and teeth. But some fossils are complete organisms or entire skeletons. This fossil dragonfly was found in fine-grained mudstone at Solnhofen, Germany.

AMAZING SURVIVALS

Although most fossils preserve only shells and bones, some show evidence of soft tissues such as skin, feathers, and even what an animal had for its last meal. The discovery of dinosaur fossils with feathers, such as this *Archaeopteryx*, has proved the link between dinosaurs and modern birds. The microscopic remains of cell structures in fossil feathers are even providing evidence of their color.



THE FOSSIL RECORD

Fossils tell us how life on Earth has evolved over many millions of years. Some types of fossils, such as marine shellfish, are very common in rocks from particular periods, and are useful "index fossils," which can be used to date rocks. Others, such as dinosaur bones, are much rarer and a whole species may be known from only a few fragments. The fossil record is not complete and has gaps. But scientists discover new fossils every day that help fill these gaps in the story of life.

◀ DELICATE WORK

A scientist in the Gobi Desert, Mongolia, cuts a fossil Protoceratops skull out of the rock that has protected it for millions of years.

